

Christianah Adekunle

Machine Learning Engineer — Speech, Applied ML & Edge Deployment

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PROFILE

Computer Science graduate (First Class Honours, CGPA 4.77/5.00) working as a Machine Learning Research Intern, currently building a Nigerian code-switching NLP pipeline. Comfortable across the full ML lifecycle — fine-tuning transformer models, evaluating with proper metrics, and shipping models to production via Docker and Hugging Face Hub rather than leaving them in notebooks. Recently fine-tuned and deployed Whisper-small for Nigerian-accented English ASR, and built a proctoring system combining real-time computer vision (YOLOv8, dlib) with client-side audio processing.

TECHNICAL SKILLS

- **ML / DL:** PyTorch, Hugging Face Transformers, scikit-learn, fine-tuning (full fine-tune on Whisper-small; familiar with LoRA/QLoRA concepts), model evaluation (WER, F1, AUC-ROC, precision/recall)
- **Speech & CV:** Whisper ASR fine-tuning, audio I/O (librosa, soundfile), YOLOv8 object/person detection, dlib facial landmark + gaze estimation, spaCy NER
- **Deployment:** Docker, Render, Hugging Face Hub, Streamlit Cloud, FastAPI, basic CI
- **Languages & Tools:** Python (primary), SQL, C (developing), Linux/bash, Git
- **Data:** Pandas, NumPy, Power BI, Excel (PivotTables, Power Query, DAX)

KEY PROJECTS

Whisper-Small Fine-Tune — Nigerian-Accented English ASR — *PyTorch, Transformers, Hugging Face Hub*

- Fine-tuned openai/whisper-small on a Nigerian-accented English speech dataset; reduced normalized WER from a pretrained baseline down to 5.25% (7.97% raw) on held-out eval data.
- Published the model and a full model card with training hyperparameters and before/after metrics to Hugging Face Hub: huggingface.co/AllehellA/whisper-small-nigerian-en

Testa — AI-Proctored Exam Platform — *FastAPI, React/Vite, YOLOv8, dlib*

- Built the ML-driven proctoring core: identity verification, blink-based liveness detection, gaze/head-pose monitoring via dlib landmarks + solvePnP, and YOLOv8n object/multi-person detection running against a live camera feed.
- Added client-side voice-activity detection and a rule-based risk scoring system that logs timestamped violations with snapshot evidence.
- github.com/i-am-christy/testa · Live: testa-blush-nu.vercel.app

Malaria Prevalence Prediction Across Nigerian States — *scikit-learn, SMOTE, Streamlit*

- Built an end-to-end ML pipeline on Nigeria Malaria Indicator Survey (NMIS) 2021 data (70,428 records); compared Logistic Regression, Decision Tree, and Random Forest with 10-fold stratified CV and SMOTE oversampling.
- Deployed the best model (Random Forest, 0.799 AUC-ROC) as an interactive Streamlit app with a household-level risk predictor and a state-level choropleth risk map.
- github.com/i-am-christy/malaria-prediction

Lassa Fever Tweet Surveillance — Dockerized Deployment — *FastAPI, Docker, Render, spaCy*

- Took a trained tweet-relevance classifier from notebook to a containerized FastAPI service deployed on Render — first end-to-end experience packaging model artifacts and dependencies into a working Docker image.
- github.com/i-am-christy/lassa_fever_surveillance_model

EXPERIENCE

Machine Learning Research Intern — VeeNode Technologies

2025 – Present

- Building a Nigerian code-switching NLP pipeline (English, Nigerian Pidgin, Yoruba, Hausa, Igbo) starting from the MENYO-20k dataset, including synthetic data generation strategies, under a Lagos AI research team.

Data Analyst Intern — Ondo State Contributory Health Commission

Jul 2024 – Nov 2024

- Built Power BI and Excel dashboards for departmental KPI reporting; used Python (Pandas) to automate data aggregation across multiple sources.

EMBEDDED & HARDWARE EXPOSURE

Covered embedded systems fundamentals in a university course (term paper project: sensor-driven waste-sorting bin concept, ESP32-based). Independently practiced circuit simulation in Wokwi to get hands-on with the platform and C — still building fluency in both. Have not yet worked with physical boards; comfortable saying so rather than overstating it.

EDUCATION

B.Tech. Computer Science — First Class Honours — Federal University of Technology, Akure (FUTA) Nov 2019 – Nov 2025

CGPA: 4.77 / 5.00